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Deep Photo Style Transfer

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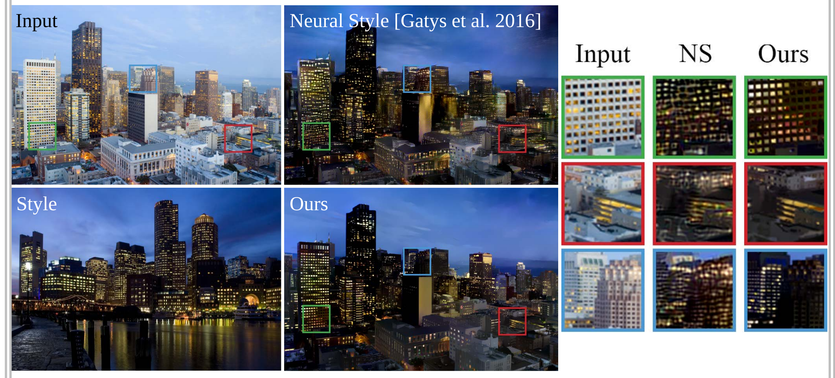
Problem Statement. Given input image I, reference style image S, output image O should have: content of I, style of S, and O should be photorealistic



Challenges

- Gram-matrix based style term [Gatys16] is global and causes spillovers due to semantic mismatch
- Neural Style [Gatys16] (that matches the statistics of VGG-19) introduces distortions that make the output look like a painting

Contributions

- Semantic segmentation to avoid content mismatch $\mathcal{L}_{s+}$
- Locally affine model as a photorealism regularization $\mathcal{L}_m$

$$\mathcal{L}_{total} = \underbrace{\left(\sum_{l=1}^{L} \alpha_l \mathcal{L}_c^l + \Gamma \sum_{l=1}^{L} \beta_l \mathcal{L}_{s+}^l\right)}_{\text{Neural Style with semantic segmentation}} + \underbrace{\gamma \mathcal{L}_m}_{\text{Photorealism regularization}}$$

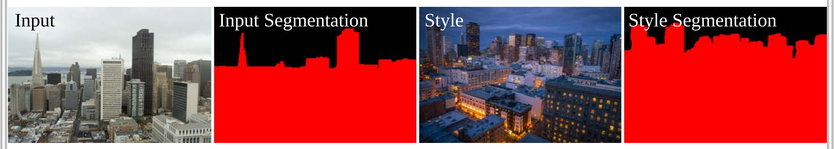
Ideas That Did Not Work

- L2 norm of color gradients
- Neural Style + local color transfer [Shih et al. 2013]
- Neural Style + multiscale gain maps [Shih et al. 2014]
- Multiscale gain maps as regularizer
- Band-limited multiscale gain maps as regularizer
- Non-overlapping local affine patches in the optimization

For more info, please check out supplemental materials.

Project Page

Semantic Segmentation



- Augment style loss with segmentation mask channels

$$\mathcal{L}_{s+}^l = \sum_{c=1}^{C} \frac{1}{2N_{l,c}^2} \sum_{i,j} \left(G_{l,c}[O] - G_{l,c}[S]\right)_{i,j}^2$$

$$F_{l,c}[O] = F_l[O]M_{l,c}[I], \qquad F_{l,c}[S] = F_l[S]M_{l,c}[S], \qquad G_{l,c}[\bullet] = F_{l,c}[\bullet]F_{l,c}[\bullet]^T$$

Photorealism Regularization

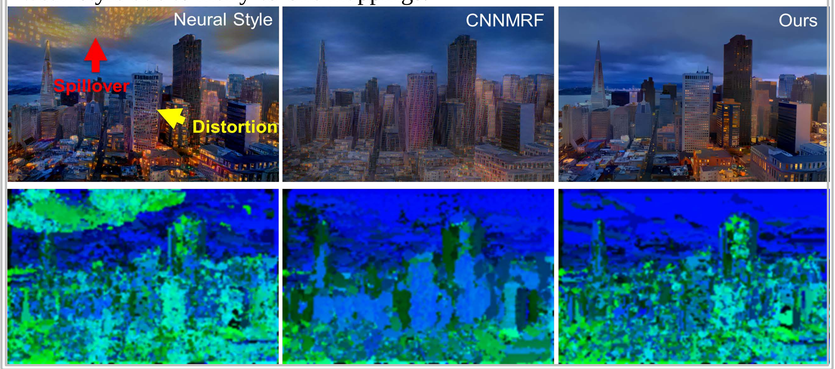
- Based on the Matting Laplacian matrix $\mathcal{M}_I$ of [Levin08]
- Penalizes outputs that are not well explained by a locally affine transform

$$\mathcal{L}_m = \sum_{c=1}^{3} V_c[O]^T \mathcal{M}_I V_c[O], \qquad \frac{d\mathcal{L}_m}{dV_c[O]} = 2\mathcal{M}_I V_c[O]$$

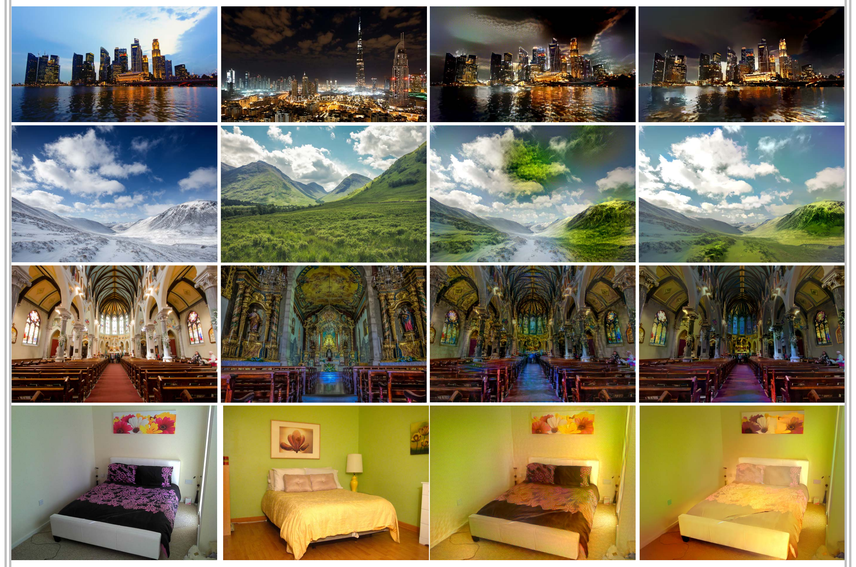
Putting Things Together

- Semantic segmentation addresses content-mismatch spillovers
- Photorealism regularization removes distortion

Our correspondence is both diverse and semantically correct , compared to Neural Style [Gatys16], which has spillovers, and CNNMRF [Li16] which has weak style due to many-to-one mappings.



Results



Input | Style | Neural Style | Ours

Input | Style | Photoshop MatchColor | Ours

User Studies. Our deep photo style transfer is preferred for photorealism compared to [Gatys16] and [Li16], and preferred for style faithfulness 86.8% over histogram transfer (5.8%), statistics transfer (4.6%) and Photoshop MatchColor (2.9%).

Acknowledgement

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References

1. *[Levin08]. "A closed-form solution to natural image matting"*
2. *[Shih13]. "Data-driven Hallucination of Different Times of Day from a Single Outdoor Photo"*
3. *[Shih14]. "Style transfer for headshot portraits"*
4. *[Gatys16]. "Image style transfer using convolutional neural networks"*
5. *[Li16]. "Combining markov random fields and convolutional neural networks for image synthesis"*